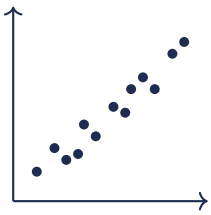
Scatter Plots

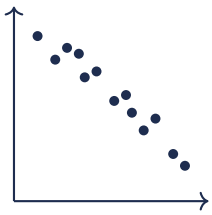
NC.8.SP.1 • NC.8.SP.2 • NC.8.SP.3

1 Association: Do the Points Make a Pattern?



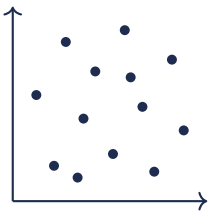
Positive linear

x up, *y* up



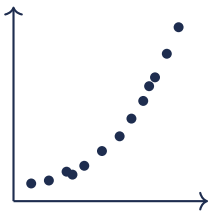
Negative linear

x up, *y* down



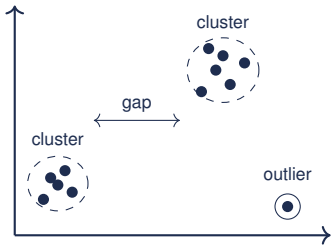
No association

no pattern



Nonlinear

curves



Cluster: points bunched together.

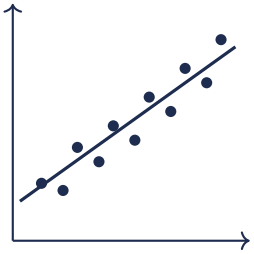
Gap: a big empty space with no data.

Outlier: a point far away from the rest.

Describe it: "As *x* increases, *y* tends to increase / decrease."
Say: **positive** / **negative** / **none** + **linear** / **nonlinear** + any clusters or outliers.

2 Line of Best Fit

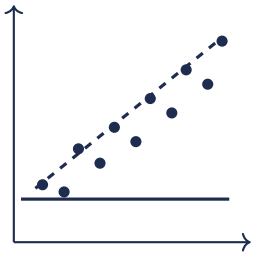
✓ **Good fit**



Follows the trend.

About half above, half below.

× **Bad fit**



Flat line: wrong direction.

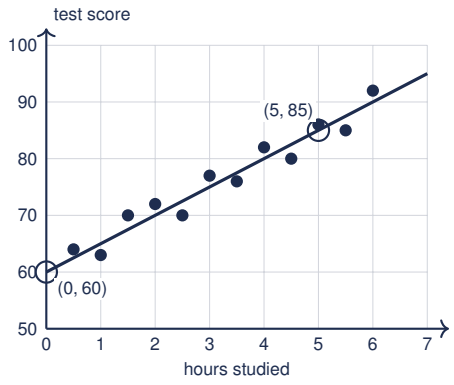
Dot to dot: just connects the ends.

How to draw it

1. Use a straight edge.
2. Follow the direction of the points.
3. About the same number of points above and below.
4. It does **not** have to touch any point.

Points close to the line → strong.
Points spread out → weak.

3 Use the Line: Equation, Meaning, Prediction



Pick 2 points **on the line** (not data dots).

Write the equation

1. $m = \frac{85 - 60}{5 - 0} = \frac{25}{5} = 5$
2. $b = 60$ (where $x = 0$)
3. $\boxed{y = 5x + 60}$

Predict: 3 hours

$y = 5(3) + 60 = 75$

About a score of **75**.

Explain it in context

Slope 5**:** the score goes up about **5 points per hour** studied.

***y*-intercept** 60**:** a student who studies **0 hours** scores about **60**.

Use units! "per" → slope. "start" or "0" → *y*-intercept.

Two-Way Tables

NC.8.SP.4 • Association in categorical data

4 Read a Two-Way Table

Survey: Do you play a sport? (100 students)

	Sport	No Sport	Total
8th grade	36	24	60
7th grade	16	24	40
Total	52	48	**100**

Two categories about the **same** people:
grade and sport.

Inside cells: how many fit **both**.
36 students are 8th graders **and** play a sport.

Totals: add across a row or down a column.
36 + 24 = 60 36 + 16 = 52

Grand total: everyone, 100.

5 Relative Frequency: Change Counts to Percents

Relative frequency = part ÷ total. Which total? Decide first.

Row relative frequency (divide by the **row** total)

	Sport	No Sport	Total
8th	$\frac{36}{60} = 0.6 = \mathbf{60}\%$	$\frac{24}{60} = \mathbf{40}\%$	100%
7th	$\frac{16}{40} = 0.4 = \mathbf{40}\%$	$\frac{24}{40} = \mathbf{60}\%$	100%

$36 \div 60$

$$60\overline{)36.0} = 0.6$$
$$-360$$
$$0$$

$16 \div 40$

$$40\overline{)16.0} = 0.4$$
$$-160$$
$$0$$

Each **row** adds to 100%.

Type	Divide by	Answers the question
Row	row total	Of the **8th graders**, what percent play a sport? $\frac{36}{60} = 60\%$
Column	column total	Of the **sport players**, what percent are 8th graders? $\frac{36}{52} \approx 69\%$
Total (joint)	grand total	Of **all students**, what percent are 8th graders who play? $\frac{36}{100} = 36\%$

6 Is There an Association?

Compare the percents.

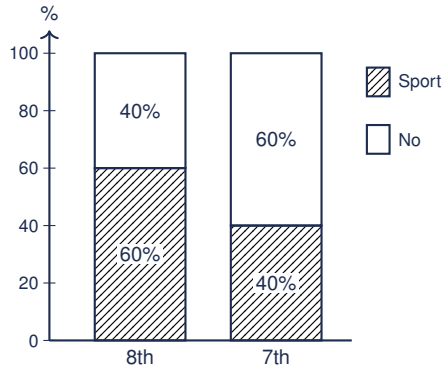
Percents are **very different** → **association**

Percents are **about the same** → **no association**

Our survey:
8th graders who play a sport: **60**%
7th graders who play a sport: **40**%

60% vs. 40% is a big difference. **Yes**, there is an association: 8th graders are more likely to play a sport.

Segmented bar graph (optional)



Bars look different → association.